Kendriya Vidyalaya Sangathan,Varanasi Region

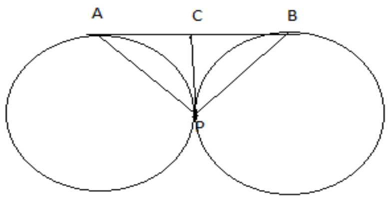
Class X Session 2024 – 25 (Marking Scheme)

MATHEMATICS STANDARD (Code No. 041)

Time : 3 hours Max Marks : 80

Q. No	Section A	Marks
1	(a) 1650	1
2	(b) 2	1
3	(b) 8 cm	1
4.	(c) $\frac{b^2}{4a}$	1
5	(a) 22 cm	1
6	(d) – 8	1
7	(d) 32.5^0	1
8	(c) – 4	1
9	(c) $2\sqrt{m^2+n^2}$ unit	1
10	(b) πd^2	1
11	(a) $S_2 - 2S_1 = d$	1
12	(d) 0	1
13	(b) 7	1
14	(a) 60^0	1
15	(c) 1/4	1
16	(c) 1/24	1
17	(b) 70	1
18	(d) – 3/7	1
19	(a). Assertion (A) and Reason (R) are True; Assertion (A) is a correct explanation for Reason (R).	1
20	(d) Assertion (A) is False, Reason (R) is True.	1
Q. No	**Section B**	**Marks**
21	For 70 – 5 = 65, 125 – 8 = 117 For prime factorization of 65 & 117 For HCF(65, 127) = 13	½ ½ x2 = 1 ½
22	For AP = AR = 5 cm & BP = PO = 4 cm For AB + BC + AC = 32 cm For 9 + 4 + x + x + 5 = 32 cm For getting x = 7 cm, BC = 11 cm, AC = 12 cm	½ ½ ½ ½
23.	For $\frac{AD}{DE} = \frac{AE}{EC}$ For $\frac{x}{x-2} = \frac{x+2}{x-1}$ For calculation and x = 4 OR For given, to prove and correct figure For AD/DB = AE/EC = 1 For DE // BC using converse of BPT	½ ½ ½ x2 = 1 ½ 1 ½
24.	For A + B = 90^0, A – B = 30^0 For calculation & getting A = 60^0 & B = 30^0	½ x 2 =1 ½ x 2 =1
25.	For $2r + l = 16.4$ cm For substituting r = 5.2 and getting $l = 6cm$ For getting the value of $\frac{\theta}{360^0} = \frac{3}{\pi r}$ using $l = 6$ For getting area of sector = 3r = 15.6 cm^2	½ ½ ½ ½
	OR	

	For expression of the area of minor segment For entries of the correct values in the expression For calculation and finding the correct area = 14 cm^2	½ ½ ½ x 2 = 1

Q. No	Section C	Marks
26.	For contrary assumption For simplification and getting 7 divides p For letting p = 7m and getting 7 divides q also For getting 7 divides p & q both. So p & q are not co-primes For conclusive statement	1 ½ ½ ½ ½
27.	For age of Mohan = x & sum of ages of his sons = y For getting equations x = 2y & x – y = 20 For solving & getting x = 40 years	½ ½ x 2 =1 1+ ½=1½
28.	For given, to prove & construction For proving the two triangles congruent using SSS or RHS For getting the corresponding angles equal (by c.pc.t) in both pairs of congruent triangles. For using straight angle / co-interior angle properties to prove the result	1 ½ ½ ½ x 2 = 1
	OR  For correct figure and construction For AC = CP and BC = CP $\angle CAP = \angle CPA = x, \angle CBP = \angle CPB = y$ $x + x + y + y = 180^0$ $2(x + y) = 180^0$ $(x + y) = 90^0$ $\angle APB = 90^0$	 1 1 1
29.	For model class 120 – 140 For $f_1 = 70, f_0 = 60$ & $f_2 = 40$, $l = 120, h = 20$ For correct formula and correct entries For calculation and correct mode = 125	½ ½ ½ x 2 = 1 ½ x 2 = 1
30.	L.H.S = $\frac{\tan A}{1+secA} - \frac{\tan A}{1-secA}$ $= \tan A \left(\frac{1}{1+secA} - \frac{1}{1-secA}\right)$ = tan A$\left(\frac{1-secA-1-secA)}{(1+secA)(1-secA)}\right)$ = tan A $\left(\frac{-2secA}{1-sec^2A}\right)$ = tanA $\left(\frac{-2secA}{-tan^2A}\right)$ = 2 . $\frac{secA}{tanA}$ = $\frac{2}{sinA} = 2\ cosecA$ = R.H.S	½ ½ ½ ½ ½ ½
	OR L.H.S = $q(p^2 - 1)$ $= (sec\theta + cosec\theta)\{(sin\theta + cos\theta)^2 - 1)\}$ $= (sec\theta + cosec\theta)(sin^2\theta + cos^2\theta + 2sin\theta cos\theta - 1)$ $= (sec\theta + cosec\theta)(1 + 2sin\theta cos\theta - 1)$ $= (sec\theta + cosec\theta)(2sin\theta cos\theta)$ $= 2sin\theta cos\theta sec\theta + 2sin\theta cos\theta cosec\theta)$	½ ½ ½ ½ ½

	$= 2\,(sin\theta + cos\theta) = 2p = R.H.S$	½
31	For getting $\alpha+\beta = \frac{14}{p}, \alpha\beta = \frac{8}{p}$ $\alpha + 6\alpha = \frac{14}{p}, 6\alpha^2 = \frac{8}{p}$ $\quad (\beta = 6\alpha)$ $7\alpha = \frac{14}{p}, 6\alpha^2 = \frac{8}{p}$ For solving and getting $p^2 - 3p = 0$ For $p = 3\,, p \neq 0$	½ x 2 = 1 ½ 1 ½
Q. No	**Section D**	**Marks**
32.	For the steps to get $2x^2 - 19x + 42 = 0$ For the steps to find the solution For correct solution $x = 6, 7/2$. OR For finding $T_1 = \frac{180}{x}, T_2 = \frac{180}{x+9}$ For $T_1 - T_2 = 1$ For substituting the values and getting $x^2 + 9x - 1620 = 0$ For steps of calculations and getting $x = 36, x = -45$ (not possible)	1 x 2 = 2 1 x 2 = 2 ½ x 2 =1 ½ x 2 = 1 ½ 1 x 2 = 2 ½ x 3 =1½
33.	For correct statement of theorem For Given, To prove & Construction and correct figure For the steps of correct Prove	1 ½ x 3 = 1½ ½ x 5 = 2½
34.	For cumulative frequency table For N/2 = 50, N= x + y + 76, Median class = 500 – 600,f = 20, cf = x +36 , h = 100, x + y = 24 , $l = 500$ For correct formula and entries of correct values For calculation and getting x = 9, y = 15	1 1 1 1 x 2 = 2
35.	For correct figure For slant height of the conical part $l = 3.5$ m For correct expression for the TSA of 100 tents = $100\pi r(l + 2H)$ For substitution of correct values For calculation and getting the correct TSA = 7590 m^2 For calculating the cost of canvas = Rs379500 OR For correct figure For cone h = 10 cm, R = 5 cm .For sphere r = ½ cm. For the expression n $\times \frac{4}{3}\pi r^3 = \frac{1}{4}\, of\, \frac{1}{3}\pi R^2 h$ For the entries of correct values For calculation For n = 125 leads	1 1 1 ½ 1 ½ 1 ½ 1 1 1 ½

Q. No	Section E	Marks
36.	(a) For applying tan 30^0 and getting, BC = $8\sqrt{3}$m = 13.84 m (b) For applying tan 45^0 and getting, AC = 24 m (c) For applying sin 45^0/ cos45^0/ Pythagoras theorem to get OC =$24\sqrt{2}$ m OR For applying tan 60^0 and getting AC = $24\sqrt{3}$m = 41.52 m	½ + ½ = 1 ½ + ½ = 1 1+ 1 = 2 1

	For increase in height of tower = 41.52 – 24 = 17.52 m	½ + ½ = 1
37.	(a) For formula for nth term and a_9 = 61 (b) For a_n = 68 & n = 10 (c) For formula for S_n and calculation and $S_{13} = 611$ OR For $S_n = 810$ For entries of correct values to get $7n^2 + 3n - 1620 = 0$ and calculation For n = 15	½ + ½ = 1 ½ + ½ = 1 ½ +1+ ½ = 2 ½ 1 ½
38.	(a) For A(2, 6) and C (8, 3) , distance formula & AC = $3\sqrt{5}$ unit (b).For mid- point formula & mid- point = (5, 4.5) , B (4,5) is not the mid- point of AC (c) For correct use of section formula & getting the ratio 1: 2 OR For a form of point on x – axis (x , 0) For correct use of distance formula $(x-2)^2 + 36 = (x-4)^2 + 25$ & calculations For point on x – axis equidistant from A & B = (1/4, 0)	½ + ½ = 1 ½ + ½ = 1 1+ 1 = 2 ½ ½ + ½ = 1 ½